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Article in *International Journal of Environmental Health Research* · May 2019

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To cite this article: Divyashree M, Deekshit Vijaya Kumar, Krishna Kumar Ballamoole, Veena Shetty A, Anirban Chakraborty & Indrani Karunasagar (2019): Occurrence of antibiotic resistance among gram-negative bacteria isolated from effluents of fish processing plants in and around Mangalore, International Journal of Environmental Health Research, DOI: [10.1080/09603123.2019.1618799](https://doi.org/10.1080/09603123.2019.1618799)

To link to this article: <https://doi.org/10.1080/09603123.2019.1618799>



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ARTICLE



Occurrence of antibiotic resistance among gram-negative bacteria isolated from effluents of fish processing plants in and around Mangalore

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ABSTRACT

The presence of antibiotic-resistant bacteria in seafood not only poses a serious health risk for the consumers but also contributes to the spread of these antibiotic-resistant bacteria in the natural environments through the effluents discharged from the fish processing plants. The aims of this study were to isolate Gram-negative bacteria from the effluents of fish processing plants in and around Mangalore, India and to profile their antibiotic resistance pattern. Maximum resistance was seen for ampicillin (40.78%) followed by tetracycline (40.22%) and nitrofurantoin (29.05%). Further, the detection of genes that contribute to antibiotic resistance revealed the presence of sulfonamide resistance genes (*su1* and *su2*) and extended spectrum β -lactamase genes (*bla*_{CTX-M} and *bla*_{TEM}) in a few isolates. The presence of such bacteria in fish processing effluents is a matter of great concern because they can contribute significantly to the antibiotic resistance in the natural environment. It is imperative that seafood processing plants follow the safe disposal of effluents in order to reduce or eliminate the antibiotic resistance menace.

ARTICLE HISTORY

Received 19 November 2018
Accepted 10 May 2019

KEYWORDS

Antibiotic resistant bacteria; fish processing plant effluents; gram negative bacteria; antibiotic resistance genes

Introduction

Antibiotics are primarily meant for therapeutic applications to treat infectious conditions in humans and animals. However, in recent years, the food production sector has witnessed unprecedented use of antibiotics, mainly as growth promoters or growth stimulators (Founou et al. 2016). The indiscriminate and non-judicious use of antibiotics coupled with the unavailability of stringent antibiotic policies has led to the emergence of antimicrobial resistant bacteria (Mazel and Davies 1999). These bacteria can reach the environment through untreated effluents from various hospitals and industries and can contaminate the surface water, groundwater, and estuaries (Sharma et al. 2010; Akin 2016). Estuaries are highly productive and serve as major aquaculture production areas. During monsoon, the surface runoffs in the estuarine environment are generally high. Since antibiotic use is quite rampant in aquaculture, it is expected that the seafood harvested from these areas are contaminated with bacteria, harboring resistance genes (Kumar et al. 2005), if not intrinsically resistant.

Fish processing plants are one of the major sources for the development and dissemination of antimicrobial resistant bacteria. It is well known that the fish processing industries use a large volume of water during initial washing, butchering and general cleaning of the processing equipment (Islam et al. 2004). These discharges are rich in organic matters and can serve as an important nutrient source for the microbial growth, since they carry fish muscle pieces, skin, scales, fins, fat, oil, protein, head, tail, shells, inedible parts (Bonsdorff et al. 1997). The bacterial load in the fish processing plants reflects the water quality in which they were caught. It also gives an idea of possible post-harvest contamination. Recently, multidrug-resistant *E. coli* have been isolated from fish effluents (Sivaraman et al. 2016), which indicates high contamination and improper treatment of fish effluents. Additionally, the use of poor quality ice and water during post-harvest processing can also contribute to contamination of the seafood.

Different genera of Gram-negative bacteria such as *Enterobacter*, *Klebsiella*, *Citrobacter*, *Serratia*, *Pseudomonas*, *Aeromonas*, *Alcaligenes*, and *Vibrio* have been isolated from fish contact surfaces. The development of antibiotic resistance has been observed in *Aeromonas* spp., *Edwardsiella* spp., and *Vibrio* spp. (DePaola et al. 2003).

Consumption of contaminated seafood may lead to the transmission of resistance genes to the human intestinal microbiota (Casals-Pascual et al. 2018). Hence, it is important to monitor the resistance in pathogenic as well as indicator bacteria associated with fish products intended for human consumption. ESBL-producing *E. coli* are usually dominant in hospital wastewaters as compared to fish processing wastewaters (Divyashree et al. 2015). However, a recent study has reported the presence of ESBL producing *E. coli* in seafood samples collected from retail fishery outlets in Veraval, Gujarat, India (Sivaraman et al. 2017). In addition, previous studies from other geographical areas have highlighted the presence of tetracycline resistance genes in bacteria isolated from aquaculture and other aquatic environments (Schmidt et al. 2001; Mirand and Zemelman 2002; Kim et al. 2005). However, reports on the presence of antibiotic-resistant bacteria in seafood processing plant effluents have been very few. The present study was an attempt to screen for antibiotic-resistant bacteria in fish processing wastes to understand the role of the processing plant effluents in increasing the level of resistant microorganisms.

Materials and methods

Sample collection and bacteria isolation

A total of 45 untreated effluent samples from various sites of the drainage system prior to the release into the sewage system were collected from two fish processing plants over a period of two years (2013–2015). The effluent samples 100 mL each were collected in sterile bottles and transported to the laboratory for microbiological analysis. To obtain the isolated colonies, the samples were serially diluted (ten-fold) in sterile physiological saline (0.85% NaCl) and 100 µl of each aliquot was spread onto the non-selective Trypticase Soya Agar (TSA) (HiMedia, Laboratories Pvt. Ltd) plates followed by incubation at 37°C for 24 h. From each plate, 10 morphologically different colonies were picked and subcultured on different selective media such as MacConkey agar, Eosin Methylene Blue (EMB) agar, Hektoen Enteric Agar (HEA), Thiosulfate Citrate Bile sucrose Agar (TCBS) and *Pseudomonas* isolation agar for bacterial identification and confirmation using a battery of biochemical tests.

Antibiogram analysis

Antibiotic susceptibility test was performed according to the standard Kirby–Bauer disc diffusion method for 12 different commercially available antibiotic discs (HiMedia, India) as per the recommendations of the Clinical and Laboratory Standards Institute (CLSI) using ATCC strains as a control. Commercially available antibiotic discs (HiMedia, Laboratories Pvt. Ltd) used in the

study were as follows: ampicillin (AMP, 10µg); cefotaxime (CTX, 30µg); ceftazidime (CAZ, 30µg); chloramphenicol (C, 30µg); cotrimoxazole (COT, 25µg); tetracycline (TET, 30µg); nitrofurantoin (NIT, 300 µg); nalidixic acid (NA, 30 µg); ciprofloxacin (CIP, 5 µg); gentamicin (GE, 10 µg); imipenem (IMP, 10 µg); and meropenem (MRP, 10 µg).

Screening for extended – spectrum β -lactamase (ESBL) production

On the basis of the antibiogram, the isolates that exhibited resistance to at least one of the cephalosporin antibiotics were presumed positive for ESBL production as per the CLSI guidelines (2012). The presumptive ESBL producers were confirmed by double disc diffusion test (DDDT) (Usha et al. 2013) and triple ESBL detection strip Ezy MIC™ (MIX+/MIX) (HiMedia, Laboratories Pvt. Ltd) as per manufacturer's instructions.

Detection of antibiotic resistance genes by PCR

Total DNA was extracted following the protocol described by (Ausubel et al. 1995). The genes responsible for conferring resistance to different classes of antibiotics like tetracycline [*tetA*, *tetB*, *tetC*, *tetD*, *tetE*, *tetG*], sulphonamide [*sul I*, *sul II*, *sul III*], chloramphenicol [*cat1*, *cat 2*, *cat 3*], aminoglycosides *aadA1* (Ma et al. 2007), ampicillin [*bla*_{TEM}] and cephalosporin [*bla*_{CTX-M}] (Colomer-Lluch et al. 2011) were amplified by PCR using specific primers (Table 1). PCR was carried out in 30 µl reaction mixture containing 10X buffer (100 mM Tris-HCl, pH 8.3, 20 mM MgCl₂, 500 mM KCl, 0.1% gelatin), 200 mM of dNTPs (dATP, dTTP, dGTP, dCTP), 10 pmol each of forward and reverse primer and 1.0 unit of *Taq* DNA polymerase enzyme (Genie,

Table 1. Oligonucleotide primers used for the detection of antibiotic resistance genes.

Resistance Genes	Corresponding antibiotics	Oligonucleotide sequences(5'-3')	Amplicon Size (bp)
<i>tet A</i>	Tetracyclines	F: TTGGCATTCTGCATTCACCTC R: TATAGCTTGCCGGAAGTCG	494
<i>tet B</i>		F: CAGTGCTGTTGTTGTCAATTAA R: GCTTGAATACTGAGTGTTAA	571
<i>tet C</i>		F: CTTGAGAGCCTTCAACCCAG R: ATGGTCGTCATCTACCTGCC	418
<i>tet D</i>		F: GCAAACCATACGGCATTCT R: GATAAGCTGCGCGTAAAAA	546
<i>tet E</i>		F: TATTAACGGGCTGGCATTTTC R: AGCTGTCAGGTGGGTCAAAC	544
<i>tet G</i>		F: GCTCGGTGGTATCTCTGCTC R: CAAAGCCCCTTGCTTGTAC	550
<i>sul I</i>	Sulfonamides	F: TTTCCTGACCCTGCGCTCTAT R: GTGCGGACGTAGTCAGCGCCA	425
<i>sul II</i>		F: CCTGTTTCGTCCGACACAGA R: GAAGCGCAGCCGAATTCAT	435
<i>sul III</i>		F: ATGAGCAAGATTTTGGAAATCGT R: CTAACCTAGGGCTTTGGATATT	792
<i>cat 1</i>	Chloramphenicol	F: AACCAGACCGTTACAGCTGGAT R: CCTGCCACTCATCGCAGTAC	549
<i>cat 2</i>		F: AACGGCATGATGAACCTGAA R: ATCCCAATGGCATCGTAAAG	547
<i>cat 3</i>		F: ATCGGCATCGTTACCATGT R: ATCCCCTTCTTGCTGATATT	531
<i>aadA1</i>	Gentamicin	F: TTTGCTGTTACGGTGAC R: GCTCCATTGCCAGTCG	497
<i>bla</i> _{TEM}	Ampicillin	F: TCACCCAGAAACGCTGGTG R: TCCGCCTCCATCCAGCTCA	569
<i>bla</i> _{CTX}	Cefotaxime	F: ACGTTAAACACCGCCATTCC R: CGGTGACGATTTAGCCGC	356

Bangalore). Amplification was carried out in a thermocycler (BioRad, USA) with the optimized PCR program (Table 1). The amplified products were resolved in 1.5% agarose gel and observed under Gel Documentation system (Bio-Rad, USA).

Results

In this study, a total of 179 Gram-negative bacteria were isolated from the effluents of two fish processing plants. The isolates comprised 56 *Proteus* spp. (30.72%), 31 *E. coli* (17.31%), 23 *Pseudomonas* spp. (12.84%), 16 *Aeromonas* (8.93%), 15 *Citrobacter* spp. (8.37%), 11 *Enterobacter* spp. (6.14%), 11 *Salmonella* spp. (6.14%), 7 *Vibrio* spp. (3.91%), 4 *Shigella* spp. (2.23%), 3 *Klebsiella* spp. (1.67%), and 2 *Morganella* spp. (1.11%). The genus level bacterial identification was done by conventional methods using a battery of biochemical tests.

Antibiotic resistance profiling

The antibiogram of the isolates revealed that 23.46% of the isolates were multidrug resistant. Maximum resistance was seen for ampicillin at 40.78% (73/179), followed by tetracycline (40.22%; 72/179), nitrofurantoin (29.05%; 52/179), nalidixic acid (18.44%; 33/179), and cefotaxime (8.94%; 16/179). Interestingly, all the isolates were sensitive to ciprofloxacin. The graphical details of the resistance pattern are shown in Figure 1

Phenotypic detection of ESBL producers

Of the 179 isolates, 16 (8.93%) and 13 (7.26%) isolates showed resistance to cefotaxime and ceftazidime, respectively, while 10 (17.9%) were resistant to both the cephalosporins. Four isolates which comprised one each of *E. coli*, *Proteus* spp., *Aeromonas* spp. and *Enterobacter* spp. were found to be ESBL positive by confirmatory tests.

Detection of antibiotic resistance genes by PCR

Of 72 tetracycline resistant isolates tested, the presence of *tetD* gene was observed in two isolates. Among 13 co-trimoxazole resistant isolates, 2 isolates harbored *sul I* and *sul II* genes and none carried the *sul III* gene. None of the isolates harbored chloramphenicol resistance genes (*cat I*, *cat*

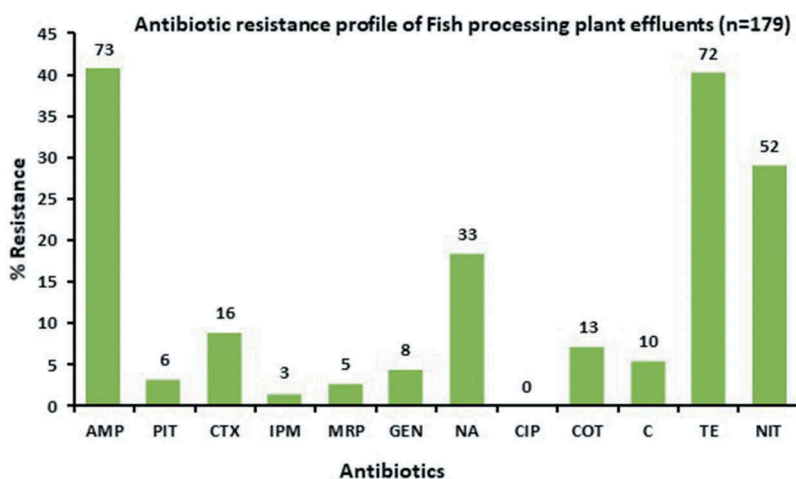


Figure 1. Antibiotic resistance profile of fish processing plant effluent isolates.

2 and cat 3) and aminoglycoside (*aadA1*) resistance genes. Among the four phenotypically confirmed ESBL isolates, two isolates showed the presence of *bla*_{CTX-M} and *bla*_{TEM} gene, while the remaining two isolates did not carry any beta-lactamase resistance genes. Interestingly, the majority of the isolates were negative for the presence of antibiotic resistance genes although they were phenotypically resistant.

Discussion

Fish processing plants require a large volume of water for cleaning and washing of the produce and the equipment. Thus, the amount of wastewater that gets discharged as effluents to the surrounding environment from the fish processing plants is considerably high. In this study, the isolates obtained from fish processing effluents comprised several *Enterobacteriaceae* members. Among them, *Proteus* (30.72%) was found to be the most frequently isolated bacteria, and this observation corroborates with the study of Chen et al. (2005) on finfish and shellfish. Screening of seafood for the presence of *E. coli* is considered a gold standard for assessing the fecal contamination of seafood in India and other countries (Jeyasekaran et al. 1990). In this study, the presence of *E. coli* (17.31%) among the isolated bacteria indicated fecal contamination, which could be due to the use of poor quality water or ice during the processing of fishes. The other source of contamination may include personal hygiene during unloading the fishes, washing and separating ice, sorting grading and re-icing before shipping (Montville et al. 2002). Indeed, a previous study has reported the presence of total and fecal *E. coli* in fish effluents, at levels higher than the recommended standards (Sivaraman et al. 2016). The presence of other pathogens such as *Salmonella* and *Vibrio* spp. in seafood from Mangalore has been well documented (Dileep et al. 2003; Deepanjali et al. 2005; Wogu et al. 2010; Kumar et al. 2011). Although the presence of *Salmonella* spp. (6.14%) and *Vibrio* spp. (3.91%) was relatively less in this study, the mere presence of these pathogens suggests that there is a potential risk of seafood-borne illnesses among the community.

In this study, 23.46% of the isolates were multidrug resistant, with high resistance to ampicillin (40.78%). This data is consistent with the observations of several previous studies that reported a similar pattern of resistance to ampicillin in aquatic samples (Ho et al. 2000; Schmidt et al. 2000; Hatha et al. 2005). The observation of high ampicillin resistance in the present study points towards its widespread use in diverse fields such as human medicine, veterinary, aquaculture, etc. The percentage of isolates resistant to tetracycline (40.22%), was also high indicating its intensive use in fish farming. Tetracyclines in aquaculture are administered through feed or direct application in pond water for prophylaxis and treatment of diseases (Hatha et al. 2005). Our observations further highlight its impact on the emergence of resistance in the microbial population, most likely through the release of wastewater into open estuaries, as shown in a previous study on drug resistance among seafood-associated bacterial isolates from Mangalore (Kumar et al. 2014). The resistance to cefotaxime (8.94%), imipenem (1.68%) and meropenem (2.79%) observed in the isolates of the current study should be viewed seriously as these antibiotics are classified under the category of lifesaving drugs used in treating serious infections (Jaskaran and Shukla 2015). Interestingly, none of the fish processing effluent isolates were resistant to ciprofloxacin, which could be due to the restricted use of quinolones in aquaculture practices and animal husbandry. Though chloramphenicol use in aquaculture is limited (Banrie et al. 2013), 5.59% of isolates in this study showed resistance to chloramphenicol. Taken together, the results of this study show that the discharge of wastes from the fish and shrimp processing industries carrying antibiotic-resistant bacteria may pollute the coastal and nearshore environment and contribute to their spread through seafood. Indeed, it has been suggested that seafood could be an international vehicle of antibiotic-resistant human pathogens (Duran and Marshall 2005).

The fish processing plants included in the present study use the fish harvested from the coastal environment that receives untreated or partially treated sewage from the townships including

hospital, veterinary industries, and food processing units. It is well known that tetracycline, sulphonamide, chloramphenicol, and beta-lactams are extensively used in humans, veterinary, and agriculture. Hence, there was a need for detection of both phenotypic and genotypic resistance among the gram-negative bacteria in the present study. Fish has been recently reported as a reservoir of ESBL producing *E. coli* (Elhadi and Alsamman et al. 2015; Singh et al. 2017). The presence of ESBL *E. coli* isolates in the fish and fishery products at retail fishery outlets of Veraval region (Gujarat) suggest possible post-harvest contamination most likely from infected handlers, uncleaned vessels and repeated use of contaminated water in the fishery outlets (Sivaraman et al. 2017). However, there have been no reports on the prevalence of ESBL producing isolates in fish processing plant effluents to date. In this study, we found four phenotypically confirmed ESBL producing isolates from fish processing effluents, of which, two harbored the *bla*_{CTX-M} and *bla*_{TEM} genes. Despite the small number, the presence of these plasmid-associated resistance genes in fish processing effluents is a matter of concern since it spreads horizontally and poses a potential health risk to human consumers (Manjusha and Sarita 2011).

The present study investigated antibiotic resistance genes exclusively in isolates that showed corresponding phenotypic resistance for commonly used antibiotics in different fields. However, the susceptible isolates should also be screened for the presence of the silent antimicrobial resistance genes. A study conducted in Mangalore showed the presence of silent antibiotic resistance genes in phenotypically sensitive isolates which could pose a future threat to the antibiotic therapy (Deekshit et al. 2012). These silent genes if carried by conjugative plasmids can spread to other bacteria under the selection pressure of antibiotic use (Yaqoob et al. 2007). Therefore, the search for antibiotic resistance genes should not be limited to the isolates which exhibit phenotypic resistance towards a particular or a group of antibiotics but should also to be extended to susceptible isolates (Ryu et al. 2012). In our study, among the ESBLs, a small percentage of isolates showed negative for the *bla*_{CTX} and *bla*_{TEM}. This could be due to the presence of other variant genes implicated in ESBL production, which were not checked in the present study.

Of 72 tetracycline resistant isolates, only two harbored *tetD* among the genes tested. The absence of tetracycline resistance genes in some of the resistant isolates could be due to the presence of other *tet* genes, which were not included in this study or it could be due to other resistance mechanisms such as efflux-mediated resistance. It is possible that the presence of a single tetracycline resistance gene is enough to cause phenotypic resistance among the bacterial isolates (Jun et al. 2010; Deekshit et al. 2012). The prevalence of *Sul* genes among the isolates was less, only two harbored *sul I* and *sul II*, which was similar to a previous report (Yemisi et al. 2014).

Conclusion

To conclude, the high microbial load in the wastewater and their resistance to many of the conventional antibiotics strongly reflects the widespread use of antibiotics in seafood production and processing. Effluents containing resistant bacteria can contaminate the receiving river and other natural waterbodies if released without effective treatment. Seafood harvested from contaminated waters may be passive carriers of resistance which can affect humans when they enter the food chain. Hence, early detection, determination of antibiotic resistance pattern, and characterization of the isolates from a fish processing plant may help in identifying the risk factors involved in the transfer and spread of drug-resistant microorganisms. Thus, proper waste management practice in the processing plants will not only protect the environment but will also maintain seafood hygiene.

Acknowledgments

The authors are grateful to acknowledge the Indian Council for Medical Research for providing financial support to I.K. (project AMR/37/2011-ECD-1). The facilities provided for the study in Nitte (Deemed to be University) are gratefully acknowledged.

Conflict of interest

No conflict of interest declared

Funding

This work was supported by the Indian Council for Medical Research [AMR/37/2011-ECD-1].

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